Archeology Proposal

Title:

Digital Archaeology of Wikipedia: Analyzing Historical Changes to Event Pages

Objective:

In this project we will examine the changes in Wikipedia articles on major historical events and explore what the changes say about the process of documenting history online. We will compare revision histories for a selected group of articles which relate to wars, politics and disasters. Some of our guiding questions for the project include: What are the changes made to the content and structure of an article? At what point in time are changes made, and do they correspond to any increased interest in the subject matter? Do they provide new information or contradict existing claims?

In doing so we will look at the revision history as a form of digital archaeology where each layer gives us more insight into how the article was developed.

Motivation:

Wikipedia is a popular reference work and an ongoing record of collaborative knowledge creation. The existing articles are only one stage of such records. Previous versions will show how certain information was added, disputed, rearranged, or deleted.

Tracing all these elements directly relates to various issues of digital archaeology, including search, preservation, data cleansing, context restoration, and the analysis of past digital objects. It can be helpful since the researcher will be able to distinguish a steady story from one that has been changed many times, to determine types of sources and changes contributing to public histories, and to show how versioning saves evidence, which is easily overlooked in the most recent version.

Data:

The primary source will be Wikipedia's MediaWiki Action API. We will use the revision-history endpoints to pull metadata and article text. I plan to pick about twenty to thirty articles that cover historical events. Examples include the Vietnam War, Watergate

and the Chernobyl disaster. The final list will give a mix of event types and will have long revision histories so that we can study many changes.

For each article we will collect revision IDs, timestamps, editor identifiers where the data is publicly available, edit summaries, byte length and the revision text or content needed for comparisons. We will get the records programmatically in requests and save the raw responses locally as JSON. Then we will store the cleaned revision metadata and analysis results in CSV files or a small relational database.

Our goal is at fifty thousand revision records across the chosen pages depending on API availability and how deep each page's history goes. We will keep track of request failures and any missing fields, preserve the responses so that the work can be reproduced and note any API limits or parts of the history that are not available.

Potential challenges include. Suppressed revisions, anonymous edits, bot activity, inconsistent edit summaries, page moves and differences, between metadata-only and full-content responses. I will not treat editor identity or edit summaries as proof of motivation.

Approach:

First we will set inclusion criteria for event pages and test API retrieval on a sample. Next we will create a data collection script that logs errors and stores raw responses.

We will. Normalize timestamps, page identifiers, revision metadata and editor categories while keeping a link from each processed record to its raw source. We will then rebuild article development by comparing selected revisions and measuring edit frequency, article length, additions and removals changes in section structure reference-related changes where possible revert patterns.

To keep text analysis we will compare revisions at regular time intervals and around unusually large edits instead of manually reading every revision. We will combine measures with close readings of representative change periods to recover context that counts alone cannot offer.

Python will be used for API collection, data cleaning, text diffs and analysis. Pandas and standard text-processing tools will support tabulation and comparisons.

Results will be presented through charts and a short written interpretation. We will validate a sample of detected changes, against the source revision pages.

Team Responsibilities:

Jackson — Data Collection and Technical Development

Jackson will lead the technical implementation of the project, including using the MediaWiki API to retrieve Wikipedia revision histories. He will develop and test the data collection scripts, save the retrieved records, and help clean and organize the data for analysis. He will also assist with analyzing revision changes and preparing visualizations.

Chase — Data Cleaning and Organization

Chase will help select the Wikipedia articles for the project and focus on cleaning, organizing, and validating the collected revision data. He will identify missing or inconsistent records, help ensure the dataset is suitable for analysis, and assist with documenting the data collection and cleaning process.

Pablo — Revision Analysis and Research

Pablo will focus on examining Wikipedia revision histories to identify patterns in how historical-event articles change over time. He will help compare revisions, investigate significant edits, and interpret changes in article content, structure, and sourcing. He will also contribute to connecting the findings to the project's digital archaeology objectives.

Jordan — Visualization and Final Presentation

Jordan will lead the development of graphs and visualizations that communicate the project's findings. He will work with the team to determine which patterns are most meaningful to present, help synthesize the results, and coordinate the preparation of presentation slides and supporting materials. He will also contribute to the final written report.

All four team members will collaborate on selecting the articles, reviewing the findings, completing progress checks, and preparing the final presentation and report.

Timeline and Milestones:

10/20/26 - Group met to discuss our project idea and set up communication.(completed)

10/22/26 - Project Proposal (completed)

10/27/26 - Progress Check 1

10/30/26 - Select Wikipedia articles and begin collecting revision history data using the MediaWiki API.

11/6/26 - Finish collecting data and begin cleaning and organizing it for analysis.

11/10/26 - Progress Check 2

11/17/26 - Analyze revision histories, identify patterns, and create graphs to show our findings.

11/21/26 - Finalize results and prepare slides for the project presentation.

11/24/26 - Final Project Presentation

12/1/26 - Final Project Report

Expected Outcomes:

We expect to find patterns in how Wikipedia articles about historical events change over time. By looking at revision histories, we hope to see when articles receive the most edits, what types of changes are made, and how the information develops over time. We also want to see if certain events or new information lead to more activity on these articles.

Our final results will include graphs and charts that help show these patterns, along with a summary of what we discovered. We hope our findings will give us a better understanding of how Wikipedia articles are updated and how information about historical events changes over time.

Some of the expected results include:

- Editing patterns: Identify when articles receive the most edits and how often they are updated.
- Content changes: Examine what information is added, removed, or changed over time.
- Data visualization: Create graphs and charts to show trends in article revisions.
- Overall findings: Compare different historical articles and explain any similarities or differences we find.